In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2017

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST – III
(Paper–1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**BCD**	**ABCD**	**C**
2.	**A**	**ABD**	**D**
3.	**CD**	**AC**	**BD**
4.	**BD**	**ABC**	**AD**
5.	**C**	**A**	**BC**
6.	**D**	**AB**	**AC**
7.	**A**	**ABCD**	**BCD**
8.	**C**	**ABC**	**ACD**
9.	**D**	**AC**	**B**
10.	**AC**	**BCD**	**B**
1.	**7**	**3**	**3**
2.	**8**	**4**	**8**
3.	**5**	**3**	**2**
4.	**2**	**9**	**2**
5.	**6**	**6**	**3**
6.	**6**	**9**	**7**
7.	**2**	**5**	**2**
8.	**6**	**3**	**6**
9.	**4**	**2**	**1**
10.	**7**	**5**	**4**

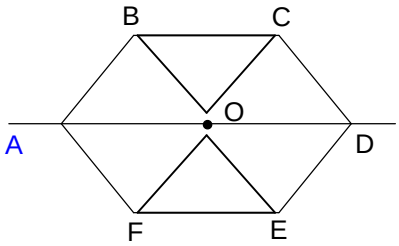
Physics

PART – I

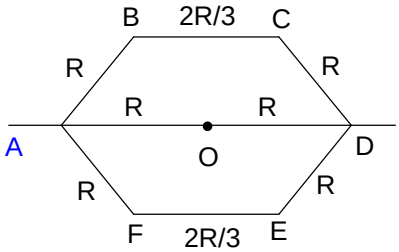
SECTION – A

1. BCD

Thermal resistance of each section $R = \frac{\ell}{KA}$



$R_{eq} = \frac{4R}{5}$

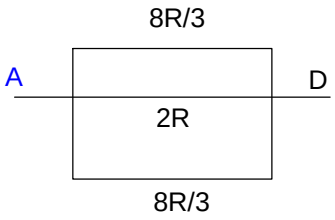


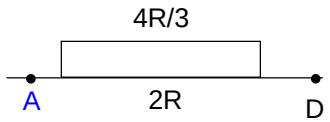
Hence $P = \frac{T_A - T_P}{4R} \times 5$

$T_D = T_A - \frac{4}{5}\frac{\ell P}{KA}$

Heat current through

ABCD & AFED = $\frac{P \times 2R}{\frac{4R}{2} + 2R} = \frac{P \times 3 \times 2}{10} = \frac{3P}{5}$





$AOD = \frac{2P}{5}$

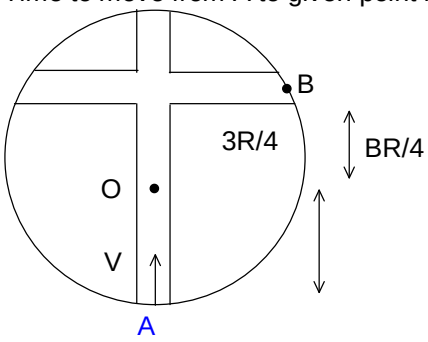
Temperature drop across AO = $\frac{2P}{5} \times R = T_A - T_0$

$T_o = T_A - \frac{2P\ell}{5KA}$

2. A

Time for B to move from B to centre of tunnel is $\frac{T}{4} = T = 2\pi\sqrt{\frac{R^3}{GM}} \quad \omega = \sqrt{\frac{GM}{R^3}}$

Time to move from A to given point is



$$\frac{1}{\omega}\left\{\sin^{-1}\left(\frac{R}{A}\right)+\sin^{-1}\left(\frac{3R}{4A}\right)\right\}=\frac{T}{4}$$

$$\sin^{-1}\left(\frac{R}{A}\right)+\sin^{-1}\left(\frac{3R}{4A}\right)=\frac{\pi}{2}$$

From above

$$\sin^{-1}\left(\frac{R}{A}\right)=\cos^{-1}\left(\frac{3R}{4A}\right)=\sin^{-1}\left(\sqrt{1-\frac{9R^2}{16A^2}}\right)$$

$$\frac{R}{A}=\sqrt{1-\frac{9R^2}{16}}$$

$$\frac{R^2}{A^2}=1-\frac{9R^2}{16A^2} \quad \frac{25}{16}\frac{R^2}{A^2}=1$$

$$A=\frac{5}{4}R$$

Hence

$$V=\omega\sqrt{A^2-R^2}=\frac{\omega\sqrt{25R^2-16R^2}}{4}$$

$$=\frac{3}{4}R\omega=\frac{3}{4}\sqrt{\frac{GM}{R}}$$

3. CD

Since fractional volume

$$\frac{V_{in}}{V_{body}}=\frac{\rho_{body}}{\rho_{liquid}}$$

$$\frac{V_{\inf}}{V_{body}}=\frac{\rho_{Bf}}{\rho_{ef}}=\frac{\rho_{Bi}.(1+\gamma_L\Delta T)}{(1+\gamma_B\Delta T)\,\rho_{\ell i}}$$

$$x_f=x_i\left(\frac{1+\gamma_L\Delta T}{1+\gamma_B\Delta T}\right)$$

with fall in temperature

$x_f > x_i$ for $\gamma_B = 3\alpha_B > \gamma_L$

4. BD

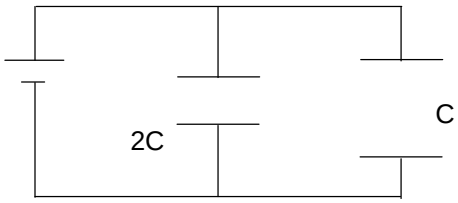
5. C

$$K.E_{max} = \frac{hc}{\lambda} - \frac{hc}{\lambda_T}$$

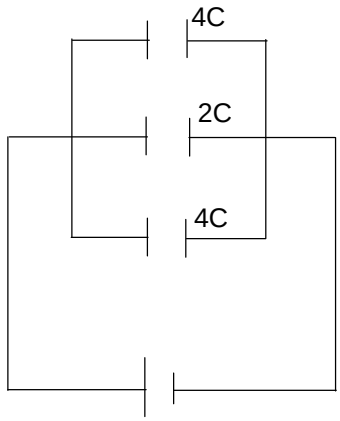
$$\ell Ed = \frac{hc}{\lambda} - \frac{hc}{\lambda_T}$$

$$\lambda_T = \left(\frac{1}{\lambda} - \frac{eEd}{hc}\right)^{-1}$$

6. D

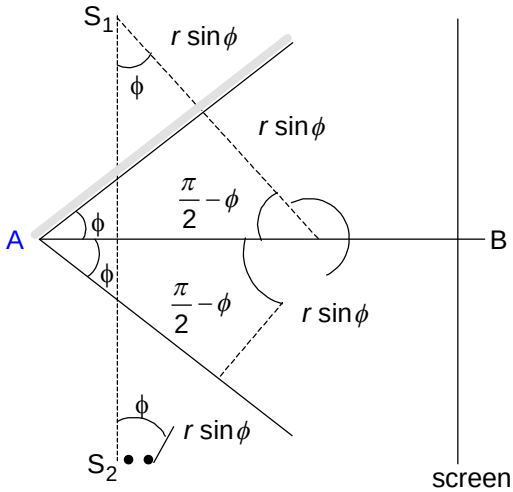


Initial
Q_{in} = 3 CV



Final
Q_f = 10 CV

7. A



A = r
SB = b

$d = S_1S_2 = 2\{2r\sin\phi.\cos\phi\} = 2r\sin 2\phi$

D = distance between source and screen

$= b + 2r\sin^2\phi$

Hence

Fringe width = $\dfrac{\left(b + 2r\sin^2\phi\right)\lambda}{2r\sin(2\phi)}$

8. C

$\tau = RC$

It do not depends on charge on capacitor

$\tau = \frac{6R}{11}.\frac{6C}{11} = \frac{36RC}{121}$

9. D

$N = m\sqrt{\left(\frac{V^2}{R}\right)^2 + g^2}$

$ma_T = -\mu m\sqrt{\left(\frac{V^2}{R}\right)^2 + g^2}$

$V\frac{dv}{ds} = -\mu\sqrt{\left(\frac{V^2}{R}\right)^2 + g^2}$

$\frac{RVdv}{\sqrt{V^4 + R^2g^2}} = -\mu_K ds$

Replacing $Z^2 = V^4 + R^2g^2$

$\int \frac{R}{2}\frac{dz}{\sqrt{z^2 - (Rg)^2}} = -\mu_K ds$

Replacing $z = Rg\sec\theta$

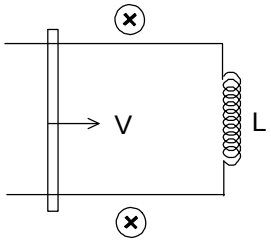
$\ln(\sec\theta + \tan\theta) = \frac{-2\mu_K}{R}ds$

$\ln\left(\sec\theta\sqrt{\sec^2\theta - 1}\right) = \frac{-2\mu_K}{R}s$

$\ln\left(\frac{z + \sqrt{z^2 - R^2g^2}}{Rg}\right) = \frac{-2\mu_K}{R}s$

$\left[\ln\left(\frac{\sqrt{V^4 + R^2g^2} + V^2}{Rg}\right)\right]_{V_0}^{0} = \frac{-2\mu_K}{R}[S]_0^s$

10. AC



$B\ell V = L\frac{dI}{dt}$

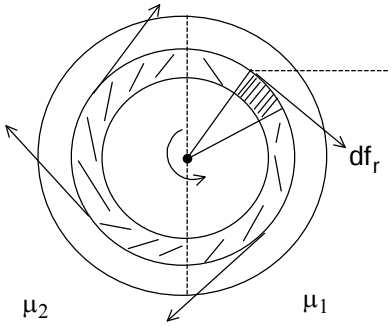
$B\ell dx = LdI$

$B\ell x = LI$

$I = \frac{B\ell x}{L}$

SECTION – C

1. 7



Let we consider a ring of radius r and thickness dr.
σ = mass per unit area.
Mass of shaded part $\sigma(rd\theta)dr$
Friction $\mu_1(\sigma rd\theta)dr$
Net horizontal force along x-axis is 0

$$\int df_y = \int(\mu_1\sigma rd\theta drg)\cos\theta$$

$$= \mu_1\sigma g\int rdr\int_{-\pi/2}^{\pi/2}\cos\theta d\theta$$

$$= 2\mu_1\sigma g\int rdr = 2\mu_1\sigma g.\frac{R}{2}\,(right\ half)\,(-ve\ y)$$

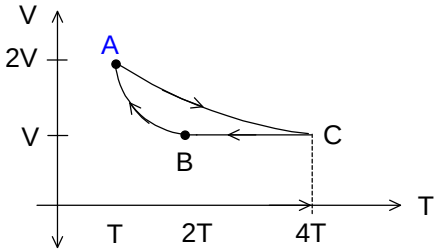
Left part

$$F_y = 2\mu_2\sigma g\frac{R}{2}\,(+ve\ y)$$

$$F_{net} = (\mu_1-\mu_2)\sigma gR^2 = \frac{(0.78-0.67).M}{\pi R}.g.R$$

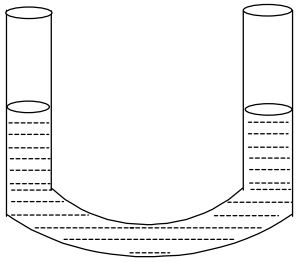
$$a = \frac{0.11\times10}{22}\times7 = \frac{7}{20}ms^{-2}$$

2. 8



AB → TV = constant = K_1
$PV^2 = K_1 \times nR$
AC → TV^2 = constant = K_2
P_{min} at A and maximum C
$P_B = 4P_A$ and $P_C = 2P_B$
Hence $\frac{P_C}{P_K} = 8$

3. 5



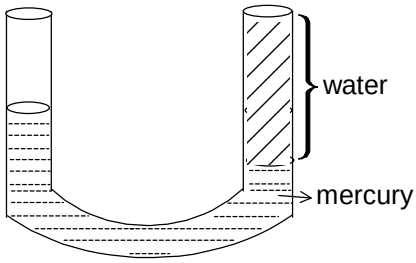
Initial

Let A = cross section area

Net force after x displacement

$-2Ax\rho_{Hg}.g = m_1 a$

$$\omega_1^2 = \frac{2A\rho_{Hg}.g}{m_1}$$



Final

Net force after displacement will be

$= -2Ax\rho_{Hg}.g = (m_1 + m_2)a$

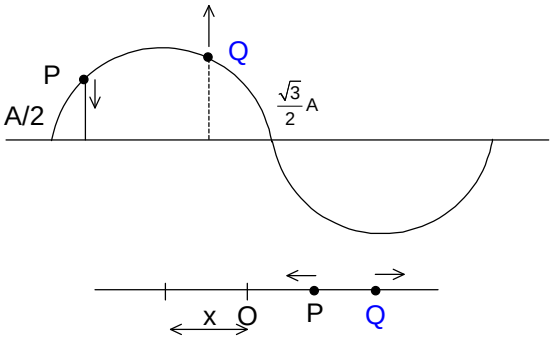
$$\omega_2^2 = \frac{2A\rho_{Hg}.g}{m_1 + m_2}$$

$$\frac{\omega_1}{\omega_2} = \sqrt{\frac{m_1 + m_2}{m_1}} = \frac{T_1}{T_2}$$

$m_1 = 80\ m$

4. 2

In case of +ve direction proceeding of wave.



In this case PQ will be horizontal if P and Q have same position propagating in opposite direction it will be in left half at position x.

For P $\frac{T}{12} + \frac{T}{4} + \frac{1}{\omega}\cos^{-1}\left(\frac{x}{A}\right)$

For Q $\frac{T}{12} + \frac{T}{4} + \frac{1}{\omega}\sin^{-1}\left(\frac{x}{A}\right)$

Equating these two

$x = \frac{A}{\sqrt{2}}$

Hence $t_1 = \frac{11T}{24}$

In case of propagating – ve x direction.

For PQ to be horizontal it will be positon i.e. right half.

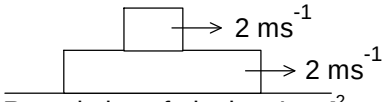
$\frac{1}{\omega}\sin^{-1}\left(\frac{x}{A}\right) - \frac{T}{12} = \frac{1}{\omega}\cos^{-1}\left(\frac{x}{A}\right) - \frac{T}{12}$

for P *for Q*

$x = \frac{A}{\sqrt{2}}$ hence $t_2 = \frac{T}{24}$

$\frac{T_1}{T_2} = 11$

5. 6



Retardation of plank = 4 ms^{-2} and retardation of block = 1.0 ms^{-2}
Hence relative acceleration of block w.r.t. plank = 3 ms
Skidding distance up to this time

$\frac{1}{2}(3)\times\left(\frac{1}{4}\right) = \frac{3}{8}\,m$

Plank is in rest now and block has velocity = $2 - 1\times\frac{1}{2} = \frac{3}{2} ms^{-1}$

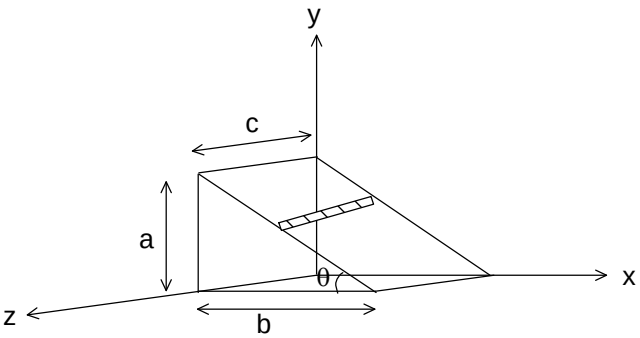
Skidding distance = $\frac{u^2}{2.a} = \frac{9}{8}m$.

Total skidding distance = $\frac{12}{8}m = 1.5\,m$

6. 6

$E = (d + x)i + E_0 i$

Net flux due to E_y = 0
Flux through left most surface is – ve and acceleration
Flux through inclined surface



$$\int (d + \ell\cos\theta) cd\ell.\sin\theta = dc\sin\theta.\ell + \frac{c.\ell^2 \sin\theta.\cos\theta}{2}$$

$$= dc.a + \frac{c.a.b}{2}$$

Net = $\frac{a.b.c}{2} = 6$

7. 2
Path difference before slits = path difference after slits.

8. 6

9. 4
e.m.f of rod at any instant
$\varepsilon_i = B\ell v$

$\theta = B\ell vc \qquad I = \frac{d\theta}{dt} = B\ell ca$

Hence net force
$2Kx - B^2\ell^2 ca = ma$

$a = \frac{2Kx}{\left(B^2\ell^2 c + m\right)} \qquad T = 2\pi\sqrt{\frac{m + B^2\ell^2 c}{2K}}$

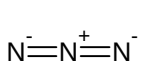
10. 7
Heat lost = heat gained
$m_i s_i \left(\theta_0 - \theta_i\right) = m_\omega L - m_\omega s_\omega \left(\theta_0 - \theta_\omega\right).$

Chemistry

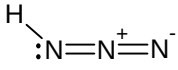
PART – II

SECTION – A

1. ABCD



Azide ion



Hydrazoic acid

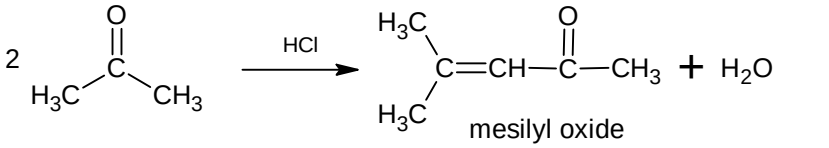
2. ABD
rate law equation is
rate = $K\left[H^+\right][HNO_2]\left[Br^-\right]$

3. AC
Correct order is
F > N > O > C > Be > B
B > Tl > Ga > Al > In

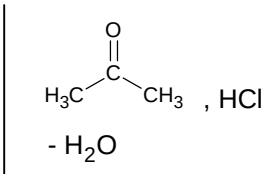
4. ABC
Facts

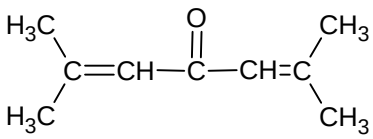
5. A
$[NiCl_4]^{2-}$ = tetrahedral, sp^3 hybridized so there are two odd e^-s in t_{2g} orbitals
$[Ni(CO)_4]$ = tetrahedral with sp^3 hybridization, no electrons unpaired

6. AB



mesilyl oxide

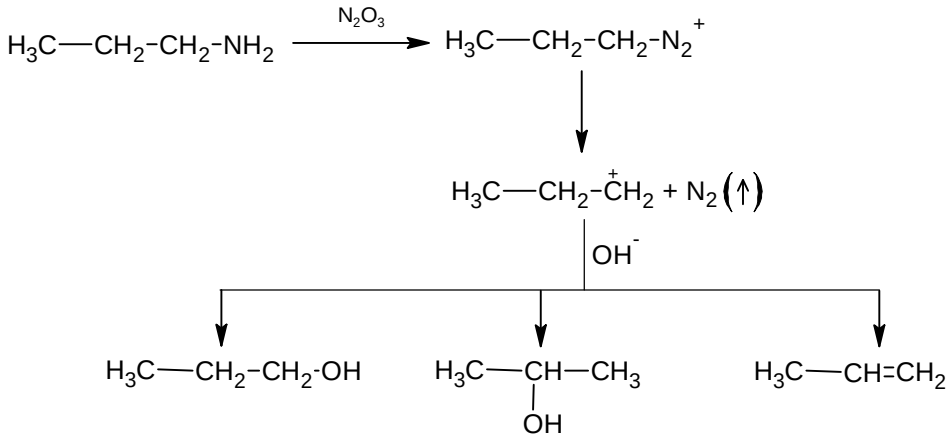




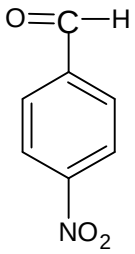
phorone

7. ABCD
Facts

8. ABC

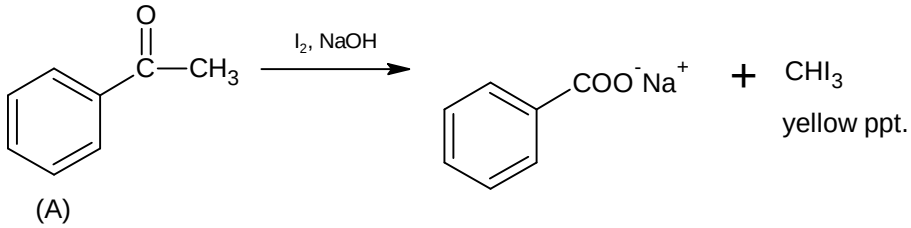


9. AC

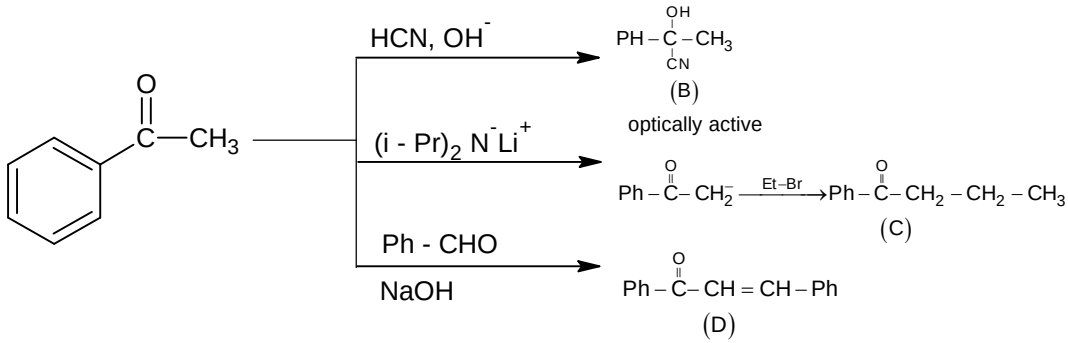


In the carbonyl carbon is more e^- deficient hence OH^- attacks over it.

10. BCD



(A)



SECTION – C

1. 3
Oxymercuration – demercuration doesn't allow rearrangement.

2. 4
(I), (II), (III) & (V) give aromatic species.

3. 3

$M^{3+} + SCN^- \rightleftharpoons MSCN^{2+}$

$[MSCN^{2+}] = \left(1.5\times10^{-3}\ M\right) - \left(1.0\times10^{-5}\ M\right) = 1.49\times10^{-3}\ M$

So, free $[M^{3+}] = \left(2.0\times10^{-3}\ M\right) - \left(1.49\times10^{-3}\ M\right)$

$= 0.51 \times 10^{-3}\ M$

$$\therefore K = \frac{\left[MSCN^{2+}\right]}{\left[M^{3+}\right]\left[SCN^-\right]} = \frac{1.49\times10^{-3}}{\left(0.51\times10^{-3}\right)\times\left(1.0\times10^{-5}\right)} = 2.92\times10^{-5} \approx 3.0\times10^{5}$$

4. 9

Mass of 25 protons = 25(1.00783 u) = 25.19575 u
And mass of 30 neutrons = 30(1.00867 u) = 30.2601
$\therefore$ Total mass = 55.45585 u
Given mass = 54.9244 u

Mass defect = 55.45585 u − 54.9244 u = 0.53145 u

$\therefore$ binding energy = $0.53145\ u\times\left(931.5\ MeV/u\right) = 495.045675\ MeV$

$$\therefore \frac{BE}{Nucleon} = \frac{495.045675}{55\ nucleons} = 9\ MeV$$

5. 6

MnO_2, Co_3O_4, Ni_2O_3, CoS, HgS, Bi_2S_3 (blackish brown)

6. 9

$$Fe^{3+} + 2H_2O \rightleftharpoons Fe(OH)^{2+} + H_3O^+$$

$$K_a = \frac{\left[Fe(OH)^{2+}\right]\left[H_3O^+\right]}{\left[Fe^{+3}\right]} = \frac{0.05\times\left[H_3O^+\right]}{0.95}$$

$$or\ 5.26\times10^{-7} = \frac{0.05\times\left[H_3O^+\right]}{0.95}$$

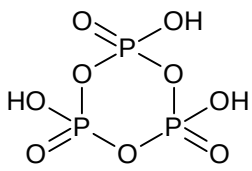
$$\therefore \left[H_3O^+\right] = 10^{-5},\ pH = 5,\ pOH = 9$$

7. 5

All Ca containing compounds impart brick red colouration to the flame . Except carnallite, alum & rock salt all other are Ca containing compounds.

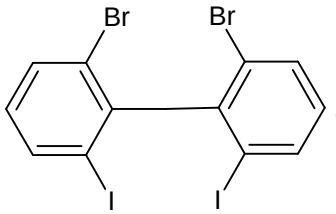
8. 3

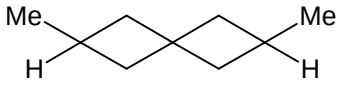
$(HPO_3)_3$

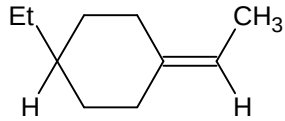


9. 2
Two stereoisomers are formed.

10. 5



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,


$CH_3CH=C=C=C=CHC_2H_5$, trans-cyclooctene are chiral.

Mathematics

PART – III

SECTION – A

1. C

$$\sum_{k=n+1}^{2n}(k-n)^2\left({}^nC_{k-n}\right)^2$$

Replace $k-n \to k$

$$\sum_{k=1}^{n}k\left({}^nC_k\right)=\sum_{k=1}^{n}\frac{k\times n!}{k!(n-k)!}$$

$$=\sum_{k=1}^{n}\frac{n(n-1)!}{(k-1)!(n-k)!}$$

$$=n\left[\frac{(n-1)!}{(k-1)!(n-k)!}\right]$$

$$\sum_{k=1}^{n}k^2\left({}^nC_k\right)^2=n^2\sum_{k=0}^{n-1}\left({}^{n-1}C_k\right)^2$$

$$=n^2\left({}^{2n-2}C_{n-1}\right)$$

2. D

${}^nC_k\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{n-k}$

Probability having exactly k male tigers

$$=P^n\,{}^nC_k\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{n-k}(n\geq k)$$

So required probability $=\sum_{n=k}^{\infty}P^n\,{}^nC_k\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{n-k}$

$$=\frac{2}{2-P}\left(\frac{P}{2-P}\right)^k$$

3. BD

$\text{adj}\,(\text{adj}A)=|A|^{n-2}A$

$|\text{adj}\,(\text{adj}A)|=|A|^{(n-1)^2}$

so $\text{adj}(\text{adj}(\text{adj}A))=|A|^{(n-2)(n-1)}(\text{adj}A)$

$|\text{adj adj}\,(\text{adj}\,A)|=|A|^{(n-1)^3}$

$=|A|^{(n-1)^3}$

4. AD

$(x-1)^2+(y-2)^2=P^2+5$

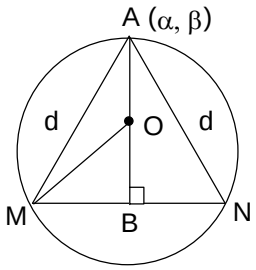
Shifted to (1, 2)

Let $a^2=P^2+5$

Equation of line MN is $\beta y+\alpha x=k$

$= mOA = \frac{\beta}{\alpha}$

$AB = a \pm \frac{k}{a}$



$(BM)^2 = (OM)^2 - (OB)^2$

$= a^2 - \frac{k^2}{a^2}$

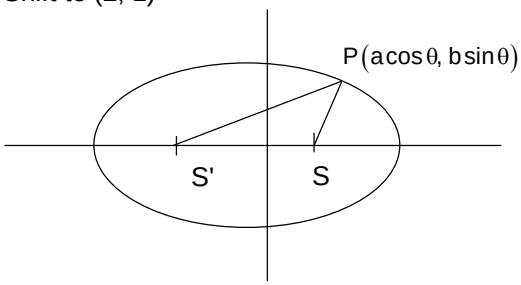
$(AM)^2 = (AB)^2 + (BM)^2$

$\Rightarrow d^2 = \left(a \pm \frac{k}{d}\right)^2 + a^2 - \frac{k^2}{d^2}$

$\Rightarrow d^2 = 2a^2 \pm 2k$

5. BC

Shift to (2, 1)



In $\Delta PSS'$

$SS' = 2ae$, $SP = a - ae\cos\theta$, $S'P = a + ae\cos\theta$

$h = ae\cos\theta$ & $k = \frac{eb\sin\theta}{e+1}$

So locus of incentres

$\frac{x^2}{a^2e^2} + \frac{y^2}{\left(\frac{be}{e+1}\right)^2} = 1$

And its eccentricity = $\sqrt{\frac{2e}{e+1}}$

6. AC

$f(x) + f(1-x) = \frac{e^x}{e^x + \sqrt{e}} + \frac{\sqrt{e}}{e^x + \sqrt{e}} = 1$

So $f_2(x) + f_2(1 - x) = 1$

$$\sum_{r=1}^{2n-1} 2f_e\left(\frac{r}{2n}\right) = 2\left[f_e\left(\frac{1}{2n}\right) + f_e\left(\frac{2}{2n}\right) + ...f_e\left(\frac{2n-1}{2n}\right)\right]$$

$$= 2\left[(n-1) + f_e\left(\frac{1}{2}\right)\right]$$

$$= 2\left[n-1+\frac{1}{2}\right]$$

$$= 2n-1$$

$$\sum_{r=1}^{1996} g\left(\frac{r}{1997}\right) = \left[g\left(\frac{1}{1997}\right) + g\left(\frac{1996}{1997}\right)\right] + \left[g\left(\frac{2}{1997}\right) + g\left(\frac{1995}{1997}\right)\right] + ...\left[g\left(\frac{998}{1997}\right) + g\left(\frac{999}{1997}\right)\right]$$

$$= 998$$

7. BCD

Use the property, foot of perpendicular from focus to tangent lies on the tangent at vertex. Using this we get equation of tangent at vertex is $6y - 4x + 1 = 0$

8. ACD

$$\frac{a+c}{2} \geq b \Rightarrow a + c > 2b$$

Similary $b + d > 2c$

$a + b + c + d > 2b + 2c$

$a + d > b + c$

and $GM \geq HM$

$$b > \frac{2ac}{a+c}$$

$$ab + bc > 2ac$$

$GM \geq HM$

$$\frac{1}{d} + \frac{1}{b} > \frac{2}{c}$$

Now $\frac{1}{a} + \frac{1}{d} > \frac{1}{b} + \frac{1}{c}$

Divide by abc, we get

$$ab + cd > 2(ac + bd - bc)$$

Divide by abcd

$$2\left(c^{-1}d^{-1} + a^{-1}b^{-1}\right) > 2\left(b^{-1}d^{-1} + a^{-1}c^{-1} - a^{-1}d^{-1}\right)$$

9. B

$$\lim_{x\to\infty} \frac{x!}{n!(x-n)!}\frac{a^n}{(x)^n}.\frac{\left(1-\frac{a}{x}\right)^x}{\left(1-\frac{a}{x}\right)^n}$$

$$\lim_{x\to\infty} \frac{a^n}{n!}\left(1-\frac{a}{x}\right)^x\left[\frac{(1)\left(1-\frac{1}{x}\right)\left(1-\frac{2}{x}\right)...\left(1+\frac{1-n}{x}\right)}{\left(1-\frac{a}{x}\right)^n}\right]$$

$$\lim_{x\to\infty} \frac{a^n}{n!} \times \frac{e^{-a}\times 1}{1^n} = \frac{a^n.e^{-a}}{n!}$$

10. B

Numbers of permutation with 2 in first position = 19!

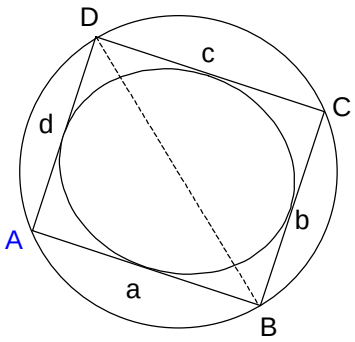
For second position = $10 \times 18!$

Similarly = $\dfrac{19! + 10 \times 18! + 10 \times 9 \times 17! + ... 10! \times 9!}{20!} = 0.1$

SECTION – C

1. 3

$a + c = b + d$



$\cos A = \dfrac{a^2 + d^2 - (BD)^2}{2ad}$

$BD^2 = a^2 + d^2 - 2ad\cos A$

and in ΔBCD

$\cos C = \dfrac{b^2 + c^2 - (BD)^2}{2bc}$

$(BD)^2 = b^2 + c^2 - 2bc\cos C$

$(BD)^2 = b^2 + c^2 + 2bc\cos A$

Compare we get $\cos A = \dfrac{ad - bc}{ad + bc}$

$= \dfrac{aar^3 - arar^2}{a^2(r^3 + r^2)} = 0,\ A = \dfrac{\pi}{2}$

2. 8

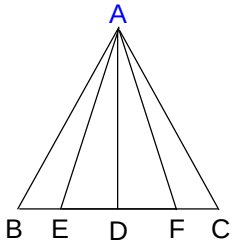
For n = 1 $\Rightarrow$ we get 16 triangle

For n = 2 $\Rightarrow$ we get 32 triangle

So for n total number of triangle is 16n

3. 2

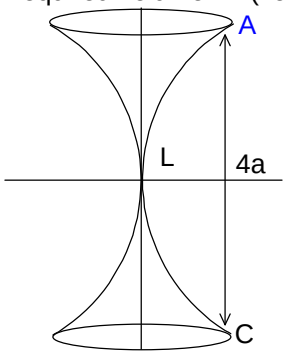
In ΔAEF



$AE^2 + AF^2 = 2(AD^2 + ED^2)$
$\Rightarrow AE^2 + AF^2 - 2AD^2 = 2$

4. 2

Required volume = 2(volume generate by the revolution about y-axis of the area ALC)



$= 2\int_0^{2a} \pi x^2 dy$

$= 2\pi \int_0^{2a} \frac{y^4}{16a^2} dy$

$= \frac{4\pi a^3}{5}$ and $a = \frac{1}{2}$

$= \frac{\pi}{10}$

so $\frac{20}{\pi}.\frac{\pi}{10} = 2$

5. 3

$\bar{v}_1 = \sqrt{a^2 - (b-1)^2}\, i + aj + \sqrt{a^2 + (b-1)^2}\, \hat{k}$

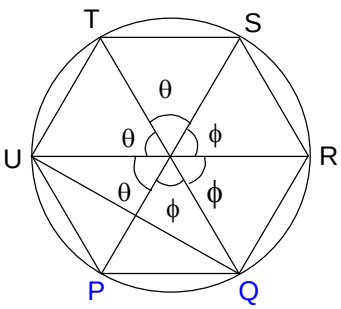
$\bar{v}_2 = \tan A i + \tan B j + \tan C k$

Let $\bar{v}_1.\bar{v}_2 = 3a$

$\sum \tan^2 A = 3\sec^2 \theta$

so min $\sum \tan^2 A = 3$

6. 7



$3(\theta + \phi) = 360^o$

$\theta + \phi = 120^o$

$\cos 120 = \dfrac{9+36-FB^2}{2(18)}$

$\cos(\theta+\phi) = \cos(120) = \dfrac{r^2+r^2-63}{2r^2}$

$\Rightarrow r^2 = 21$

$r = \sqrt{21}$

7. 2

$$I_n = \int_0^{\pi} \left(\frac{2\sin\left(n+\frac{1}{2}\right)x.\sin\frac{x}{2}}{1-\cos x} \right) dx$$

$I_n = I_{n-1} = I_{n-2} - I_0$

$I_n = I_0 \Rightarrow I_0 = \pi \Rightarrow I_n = \pi$

so $\dfrac{n\pi}{2} = \pi \Rightarrow n = 2$

8. 6

$L \equiv \dfrac{x-2}{a} = \dfrac{y-2}{b} = \dfrac{z-1}{c}$

$2a + b - 6c = 0$ & $\begin{vmatrix} a & b & c \\ 1 & -1 & 2 \\ 2 & 1 & -6 \end{vmatrix} = 0$

We get a : b : c ≡ 33 : 30 : 16

So $\dfrac{2a}{c} + \dfrac{b}{c} = \dfrac{33}{8} + \dfrac{15}{8} = 6$

9. 1

$e^x + e^{-x} + 3 - 5\cos x = 0$

$e^x + e^{-x} = 5\cos x - 3$

So only one solution x = 0

10. 4

$\sin^{-1}\left(-\dfrac{1}{2}\right) + \cos^{-1}\left(-\dfrac{1}{2}\right) = \dfrac{\pi}{2}$